

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.5: Area of Part of a Circle — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.5: Area of Part of a Circle
Driving Question	DRIVING QUESTION: How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle — like a watermelon slice at Wakulima Market — so that we can solve fair-sharing and design problems in real life?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Introducing the Circle and Its Parts	Students examined physical circular objects (bottle caps, sufuria lids, round chapati) and identified key parts: centre, radius, diameter, circumference, chord, arc, sector, and segment. Teacher should cold-call at least 4 students to label parts on a large circle drawn on the board. Allow 30-second wait time after each prompt.	A circle is a set of all points equidistant from a fixed centre. Its key measurable parts — radius (r), diameter ($d = 2r$), and circumference ($C = 2\pi r$ or πd) — form the foundation for calculating any area within the circle. Students should be able to distinguish between a sector (wedge bounded by two radii and an arc) and a segment (region between a chord and an arc).	Teachers should explain that every part of a circle is defined by its relationship to the radius. Emphasise that the watermelon slice at Wakulima Market is a real-world sector — bounded by two straight cuts (radii) and a curved rind (arc). Pedagogical note: use the analogy of cutting a round chapati — each cut from the centre outward creates a sector. This anchors the driving question from Lesson 1 and gives students a mental image to carry through all 8 lessons.
Lesson 2 Reviewing Circle Area and Introducing the Central Angle	Students reviewed the formula for the area of a full circle ($A = \pi r^2$) using concrete examples — e.g., a circular shamba plot of radius 7 m near Kericho. They then measured central angles of pre-drawn sectors using protractors and recorded the angle and the fraction it represents of 360° . Teacher should circulate and prompt: 'If the full circle is 360° , what fraction of the circle is your sector?'	The area of a full circle is $A = \pi r^2$. A central angle θ (in degrees) describes what fraction ($\theta/360$) of the full circle a sector occupies. This fraction is the critical link between the full circle area and any sector area. Students must be fluent in expressing angles as fractions of 360° before moving to the sector formula in Lesson 3.	Teachers should emphasise that 360° represents one complete revolution — a full circle. Any sector is simply a proportional slice of that full circle. Pedagogical note: draw a pie chart of a family's ugali-and-sukuma wiki meal portions on the board. Ask: 'If ugali takes up 180° of the pie chart, what fraction of the meal is it?' This proportional reasoning is the conceptual engine of the entire sub-strand. Cold-call 3 students to state the fraction for angles of 90° , 120° , and 45° before closing the lesson.

Lesson 3 Investigating Sector Area with Cut-Outs	Students cut out full circles from paper, folded or divided them into sectors of known angles (e.g., 90°, 60°, 120°), and physically compared sector areas by stacking and overlapping cut-outs. They recorded findings in a table: central angle fraction of circle predicted area calculated area. Teacher should ask: 'What do you notice when you double the angle?' Allow 45 seconds of think time before taking responses from at least 3 students.	The area of a sector is directly proportional to its central angle. Doubling the central angle exactly doubles the sector area, confirming the relationship $A = (\theta/360) \times \pi r^2$. A 180° sector has exactly half the circle's area; a 90° sector has exactly one quarter. This proportional relationship is precise and exact, not an approximation.	Teachers should use the cut-out activity results to guide students to generalise: 'If a 360° sector has area πr^2, then a θ° sector has area $(\theta/360) \times \pi r^2$.' Pedagogical note: stress that this is not a new formula to memorise in isolation — it is a direct application of proportional reasoning students already know. Connect back to the driving question: 'A watermelon slice cut at 60° from the centre — what fraction of the whole watermelon face does it represent?' Ensure students record the formula $A = (\theta/360) \times \pi r^2$ in their notes with units clearly labelled (cm², m²).
Lesson 4 Cutting Watermelon Fairly: Deriving the Area of a Sector Formula	Students worked through a structured scenario: a round watermelon of radius 15 cm is cut into slices at Wakulima Market. Each vendor cuts at a different angle. Students calculated sector areas for angles of 45°, 60°, 90°, and 120° and compared results. Teacher should pose: 'Which vendor gives the customer the biggest slice per shilling?' Cold-call 4 students to present their calculations. Require units in every answer.	Area of sector = $(\theta/360) \times \pi r^2$. A 90° sector contains exactly one-quarter of the total circle area. The formula works for any angle and any radius. Students also learned that 'fair sharing' requires equal central angles — not just equal-looking cuts — because area scales with angle. This is a key insight connecting mathematics to real-life equity and commerce at Kenyan markets.	Teachers should formalise the derivation: $(\theta/360) \times \pi r^2$ emerges because the sector is the fraction $\theta/360$ of the full circle of area πr^2. Pedagogical note: ask students to consider what happens if a market vendor cuts a 'bigger-looking' slice by increasing the radius rather than the angle — does this change the angle-based formula? This deepens understanding that both r and θ independently affect sector area. Emphasise that the formula is exact, not an estimate, when π is used symbolically.
Lesson 5 Applying the Sector Area Formula to Real-Life Problems	Students solved a set of applied problems involving sectors in Kenyan contexts: irrigated fan-shaped farm plots in the Rift Valley, a rotating sprinkler covering a 120° arc, a Nairobi roundabout garden in the shape of a sector, and a clock hand sweeping a sector in a given time. Teacher should circulate and use targeted questioning: 'What is your r? What is your θ? Have you checked your units?' Require at least one self-correction per group before submitting answers.	The formula $A = (\theta/360) \times \pi r^2$ applies directly to any real-world sector problem once the radius and central angle are correctly identified from the context. Students learned to extract r and θ from word problems, apply the formula, and express answers in appropriate units (cm², m², hectares where relevant). They also practised working with $\pi \approx 22/7$ and $\pi \approx 3.14$ and leaving answers in terms of π.	Teachers should highlight common errors: confusing radius with diameter (e.g., a watermelon described as '30 cm across' has $r = 15$ cm), and using the wrong angle (e.g., using the arc length angle instead of the central angle). Pedagogical note: display a worked example on the board using the Rift Valley irrigation scenario step-by-step, then erase selective steps and cold-call students to reconstruct the solution. This retrieval practice consolidates procedural fluency before moving to segment area in Lesson 7.
Lesson 6 Formalising the Sector Area Formula: From Watermelon Wedges to Irrigation Arcs	Students revisited their Lesson 3 and 4 findings and formally recorded the sector area formula with a derivation narrative. They then applied the formula to a design task: planning a fan-shaped irrigation arc for a maize farm in Eldoret, given fixed pipe	Learners confirmed that the area of any sector is a proportional fraction of the full circle's area: $A = (\theta/360) \times \pi r^2$. They also discovered that because r is squared in the formula, small increases in radius produce large increases in area — a non-linear	Teachers should make the r^2 effect explicit by comparing two worked examples side by side: one where θ doubles and one where r doubles. Students should see that doubling θ doubles the area (linear), while doubling r quadruples the area (quadratic).

	length (radius) and available land angle. Students sketched and labelled their sectors and computed areas. Teacher should ask: 'How does changing the pipe length by 2 m change the irrigated area?' Allow 60 seconds of paired discussion before cold-calling.	relationship. A 10% increase in radius increases sector area by approximately 21%. This has practical implications for irrigation and land-use planning.	Pedagogical note: connect this to the driving question by asking, 'If a watermelon vendor doubles the slice angle versus doubles the watermelon size, which gives the customer more area?' This is a rich discussion prompt. Ensure students update their exercise books with a fully labelled formula box before Lesson 7.
Lesson 7 Investigating Segment Area: From Sector to Slice — Unlocking the Curved Piece	Students physically cut a sector from a paper circle, then drew the chord connecting the two endpoints of the arc to create a triangle inside the sector. They cut out the triangle and placed it aside, leaving only the segment (the 'rind piece'). They recorded: Area of segment = Area of sector – Area of triangle. Teacher should ask: 'What shape is the triangle formed by the two radii and the chord?' Wait 30 seconds, then cold-call 3 students. Guide students to confirm it is an isosceles triangle with two sides equal to r.	The area of a circular segment = area of sector – area of the isosceles triangle formed by the two radii and the chord. The triangle has area = $\frac{1}{2}r^2\sin\theta$ (derived from the formula for area of a triangle using two sides and an included angle). Therefore: Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. Students must be able to identify θ as the central angle of both the sector and the triangle's included angle.	Teachers should carefully derive $\frac{1}{2}r^2\sin\theta$: the triangle has two sides of length r and included angle θ , so Area of triangle = $\frac{1}{2} \times r \times r \times \sin\theta = \frac{1}{2}r^2\sin\theta$. Pedagogical note: many students confuse the triangle's base-height formula with this sine version — emphasise that we use the sine formula here because we know two sides and the included angle, not the base and perpendicular height directly. Use a numerical example: $r = 10$ cm, $\theta = 60^\circ$, so triangle area = $\frac{1}{2} \times 100 \times \sin 60^\circ = 50 \times (\sqrt{3}/2) = 25\sqrt{3}$ cm ² . Connect to the driving question: 'The curved rind piece of a watermelon slice is a segment — now we can find its exact area.'
Lesson 8 Segment Area, Final Explanation & Fair-Share Verdict — Completing the Watermelon Puzzle	Students solved the culminating problem: a round watermelon at Wakulima Market has radius 18 cm. Three friends each take a slice cut at 80° from the centre. Using both the sector and segment formulas, they calculated (a) the area of each wedge-shaped sector slice, (b) the area of just the curved segment (the 'rind portion'), and (c) whether the three slices together cover more or less than half the watermelon face. Teacher should cold-call one student per sub-question to present to the class. Require justification for every numerical answer.	Learners derived and applied the complete formula: Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. They verified it numerically against their cut-and-subtract results from Lesson 7, confirming the algebraic and physical methods agree. They also applied the sector formula to confirm that three 80° slices cover 240° — exactly two-thirds of the watermelon face. The driving question is fully answered: exact area calculations enable fair sharing and precise design decisions in real-life Kenyan contexts.	Teachers should lead a full summary discussion connecting all 8 lessons: (1) circle parts → (2) central angles as fractions → (3–6) sector area $A = (\theta/360)\pi r^2$ → (7–8) segment area = sector – triangle = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. Pedagogical note: draw a single diagram on the board showing a watermelon slice with all labels (r, θ , sector, triangle, segment) and annotate each formula on the diagram. Ask the class: 'How would a market vendor use this to price slices fairly?' Allow open discussion for 3 minutes. Close by having students write a one-sentence answer to the driving question in their own words — this serves as both a formative assessment and a personal record of their learning journey.